Lahore University of Management Sciences

EE-5102/CS-6304 – Advanced Topics in Machine Learning

Fall 2026

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Course URL (if any)	https://lms.lums.edu.pk/

Course Basics				
Credit Hours	3			
Lecture(s)	Nbr of Lec(s) Per Week	2	Duration	75 minutes
Recitation/Lab (per week)	Nbr of Lec(s) Per Week	-	Duration	-

Course Distribution	
Core	No
Elective	Yes
Open for Student Category	Enrollment by invitation only. Please reach out to the instructor directly.
Close for Student Category	All

COURSE DESCRIPTION

This course is designed as an advanced-level offering in Deep Representation Learning (DRL) for vision and natural language, which is a subfield of machine learning. This course should not be your first formal encounter with machine learning. It builds upon the foundational understanding acquired in courses like CS331, CS535, or CS437, and aims to provide deeper insight into the core ideas, challenges, and theoretical underpinnings that drive current research in the field. The goal is to prepare students with the knowledge and tools necessary to conduct impactful, publishable research, particularly in real-world scenarios where traditional models often fall short.

While deep-learning based models have seen significant progress over the last decade, the field is currently grappling with important open questions around representation learning, generalization beyond training distributions, data and communication-efficient training, and interpretability. This course is structured around these themes, focusing not just on models, but on the principled understanding of what makes them work (or fail) in different settings. Please note:

- This course is **not** intended to introduce standard deep learning architectures (e.g., CNNs, RNNs, Transformers, GANs etc.) applied to specific modalities like audio, or vision, or language/text, nor to emphasize application-specific pipelines like object detection or sequence modeling.
- Similarly, this is **not** a seminar-style course based on in-class paper discussions. Instead, we will study the key principles and theoretical ideas that form the backbone of these advancements, and use research literature selectively to support those concepts.

COURSE PREREQUISITE(S)

- CS331/EE315: Foundations of AI and Machine Learning, or
- CS535/EE514: Machine Learning, or
- AI600: Deep learning course, or
- their equivalents at UG/GD levels.

Note: Students must have taken one of the above courses (or an equivalent) with a satisfactory grade. Prior exposure to topics like Computer Vision, NLP, or Deep Learning is helpful but not mandatory. A solid grasp of undergraduate-level Calculus, Probability, and Linear Algebra is required. Programming proficiency in Python is essential, as the course includes programming-intensive assignments. No programming tutorials will be offered, so students should be comfortable with independently implementing algorithms and models in Python.

If you're unsure whether your background is a good fit for this course, feel free to reach out or drop by during office hours for a quick discussion.

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Grading Breakup and Policy

- **Attendance (5%)**
- **Quizzes (10%):** 5
- **Homework Assignments (35%):** 5
- **Final Exam (15%):** 1

- **Final (group) Project (35%):** Students will be required to complete an independent research project in a group of 2-3 people. The projects will be research oriented and each student in the group needs to contribute significantly to algorithmic design components and implementation. The project can be on any interesting research problem related to the course that the student comes up with himself/herself or with the help of the instructor. Some project ideas will be suggested in class.

  Note that re-implementing an existing paper is not sufficient for the project. The project needs to attempt to go beyond the existing publication. The grade will depend on the project definition, how well you present them in the report, how well you position your work in the related literature, how thorough are your experiments/evaluations and how thoughtful are your analysis and conclusions. Project grades are divided as:
  - 0% for the project proposal (Mandatory)
  - 5% for the midterm report
  - 25% for the presentation, viva, and quality of obtained results, analysis, and conclusions drawn.
  - 5% for the final report in Latex template provided, along with the GitHub repository.

Examination Details	
Midterm Exam	Yes/No: No Combine Separate: Duration: Preferred Date: Exam Specifications:
Final Exam	Yes/No: Yes Combine Separate: - Duration: - Exam Specifications: 3 hours

Course Learning Outcomes	
	By the end of the course, the students should be able to:
CLO1	• Understand how to incorporate desirable properties in learning a good representation from data using mathematical principles.
CLO2	• Understand how to develop good models with limited or no labelled data.
CLO3	• Understand how to introduce efficiency and interpretability during representation learning.
CLO4	• Understand the mathematical underpinnings behind deep models.

Relation to Program Learning Outcomes

CLOs	Related PLOs	Level of Learning	Teaching Methods	CLO Attainment checked in
CLO1	PLO1: Academic Education	Cog-2	Instruction, Homework	Assignment, Project
CLO2	PLO2: Knowledge for Solving Computing Problems	Cog-3	Instruction, Homework	Assignment, Project
CLO3	PLO2: Knowledge for Solving Computing Problems	Cog-4	Instruction, Homework	Assignment, Project

Textbook(s)/Supplementary Readings

Textbook: Class material

Reference material:

- Murphy, Kevin P. ***Probabilistic Machine Learning: Advanced Topics***. MIT press, 2023.

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COURSE TOPICS			
Week	**Topics**	**Content**	**Assessment**
0-4	**Basics of Deep Representation Learning (DRL)**	**Deep Representation Learning (DRL) Basics:** o Statistical view of data and information theory measures o Representation learning and main DRL approaches — discriminative, generative and contrastive / multimodal models o Inductive biases in DRL and their role in IID vs OOD generalization. **Transfer Learning, Domain Adaptation, and Generalization**: o Transfer learning basics principles o Unsupervised domain adaptation — principles, bounds, and methods o Domain Generalization — principles, bounds, and methods o Self-supervised learning — pretext tasks and methods o OSR Problems under DG **Federated Learning**: o Federated learning basics — distributed model training in server–client setting. o Federated optimization — distributed SGD, model aggregation principles, and federated averaging (FedAvg); o Data heterogeneity challenges and methods for handling heterogeneity **Model Compression & Efficient ML**: o Model compression basics — the information bottleneck principle in deep networks, rate–distortion theory, and theoretical bounds on compression. o Network compression methods: pruning including structured vs unstructured pruning, the lottery ticket hypothesis, quantization, quantization-aware training, and straight-through estimators. o Knowledge distillation — mutual information between teacher and student representations, single-teacher distillation using logits, FitNets, and contrastive representation distillation, dark knowledge and its limits in transfer, soft labels, label smoothing regularization, decoupled KD, multi-teacher distillation.	Assignment 0 Assignment 1 Assignment 2
5-6	**Transformer Models and LLM Architectures, Training and Policy Alignment**	**LLM Architectures, Training, and Alignment:** o Transformer basics — tokens, embeddings, positional encoding, self-attention, multi-head attention, feed-forward blocks, residual connections, layer normalization, and autoregressive next-token prediction; o LLM architectures — decoder-only, causal masking, long-context transformers, sparse attention, mixture-of-experts, and multimodal extensions; o LLM training pipeline — data collection, filtering, deduplication, tokenization, pretraining objectives, optimization, scaling laws, distributed training, checkpointing, and evaluation; o Post-training — supervised instruction tuning, parameter-efficient fine-tuning, reward modeling, RLHF, PPO-based policy optimization, direct preference optimization, constitutional / AI-feedback-based alignment, safety tuning, refusal behavior, helpfulness–harmlessness–honesty trade-offs, and alignment evaluation.	Assignment 3
7-8	**Reasoning LLMs and Challenges**	**Reasoning LLMs, Families, and Challenges:** o Reasoning basics — language modeling as implicit reasoning, compositional generalization, arithmetic, symbolic, commonsense, scientific, and planning-based reasoning; o Prompting-based reasoning — chain-of-thought, few-shot reasoning, self-consistency, decomposition, scratchpads, and verifier-guided answer selection; o Search-based reasoning — tree-of-thoughts, graph-of-thoughts, best-of-N sampling, process reward models, step-wise verification, backtracking, and deliberative decoding; o Tool-augmented reasoning — code execution, calculators, retrieval, external memory, program-of-thoughts, and agentic planning;	

		○ Reasoning model families — instruction-tuned reasoners, long-CoT models, RL-trained reasoning models, verifier-based models, tool-using models, and neuro-symbolic hybrids; ○ Core challenges — reasoning faithfulness, hallucinated intermediate steps, shortcut learning, brittle generalization, overthinking, compute cost, benchmark leakage, verifier reliability, multi-step error accumulation, and evaluating reasoning beyond final-answer accuracy.	
9-10	**Test-time Adaptation of LLMs**	**Test-Time Adaptation of LLMs:** ○ Test-time adaptation basics — adapting a fixed pretrained LLM to a new input, task, user, or domain during inference without full retraining; ○ Non-parametric adaptation — in-context learning, prompt selection, prompt rewriting, retrieval-augmented context construction, memory-based adaptation, demonstration selection, and dynamic system prompts; ○ Parametric adaptation — test-time prompt tuning, adapter / LoRA updates, lightweight self-supervised updates, entropy or perplexity minimization, pseudo-labeling, and query-conditioned self-training; ○ Feedback-driven adaptation — self-correction, critique–revise loops, verifier feedback, tool feedback, and online preference signals; ○ Domain adaptation at inference — handling distribution shift, specialized terminology, private or local knowledge, and changing user constraints; ○ Challenges — instability, catastrophic forgetting, overfitting to a single query, latency overhead, privacy, safety drift, calibration, when to adapt versus retrieve, and reliable evaluation of adaptation gains.	Assignment 4
11-12	**Efficient Inference for LLMs**	**Efficient Inference for LLMs:** ○ LLM inference basics — prefill versus decode, autoregressive generation, latency, throughput, memory bandwidth, batch size, sequence length, and cost per token; ○ Memory and attention efficiency — KV caching, cache layout, paged attention, FlashAttention-style kernels, long-context serving, KV cache compression, eviction, quantization, and reuse; ○ Decoding efficiency — greedy decoding, sampling, beam search, speculative decoding, self-speculative decoding, early exiting, draft models, and acceptance-rate trade-offs; ○ Model and system efficiency — weight quantization, activation quantization, tensor parallelism, pipeline parallelism, expert parallelism, continuous batching, request scheduling, prefix caching, load balancing, and serving MoE models; ○ Deployment trade-offs — accuracy versus latency, throughput versus responsiveness, memory versus context length, batching versus interactivity, edge versus cloud inference, and efficient serving under heterogeneous workloads.	Assignment 5
13-14	**Mechanistic Interpretability in LLMs**	**Mechanistic Interpretability in LLMs:** ○ Interpretability basics — moving from black-box evaluation to understanding internal computations in transformer layers, attention heads, MLPs, embeddings, residual streams, and logits; ○ Representation analysis — probing, attribution, attention visualization, logit lens, tuned lens, activation steering, and representation geometry; ○ Causal analysis — activation patching, path patching, causal tracing, ablations, counterfactual interventions, and identifying which components cause specific behaviors; ○ Circuit analysis — induction heads, copying circuits, factual recall circuits, refusal circuits, feature interaction, and layer-wise information flow; ○ Feature-level interpretability — superposition, polysemantic neurons, sparse autoencoders, dictionary learning, monosemantic features, feature circuits, and scalable feature attribution; ○ Advanced directions — automated circuit discovery, interpretability for safety auditing, detecting deception or jailbreak mechanisms, mechanistic evaluation of alignment, limitations of interpretability methods, and the gap between local explanations and full-model understanding.	
Final Exam and Project Presentation			

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Academic Honesty

The principles of truth and honesty are recognized as fundamental to a community of teachers and students. This means that all academic work will be done by the student to whom it is assigned without unauthorized aid of any kind. Plagiarism, cheating and other forms of academic dishonesty are prohibited. Any instances of academic dishonesty in this course (intentional or unintentional) will be dealt with swiftly and severely. Potential penalties include receiving a failing grade on the assignment in question or in the course overall. For further information, students should make themselves familiar with the relevant section of the LUMS student handbook.

Harassment Policy

SSE, LUMS and particularly this class, is a harassment free zone. There is absolutely zero tolerance for any behavior that is intended, or has the expected result of making anyone uncomfortable and negatively impacts the class environment, or any individual's ability to work to the best of their potential.

In case a differently-abled student requires accommodations for fully participating in the course, students are advised to contact the instructor so that they can be facilitated accordingly.

If you think that you may be a victim of harassment, or if you have observed any harassment occurring in the purview of this class, please reach out and speak to me. If you are a victim, I strongly encourage you to reach out to the Office of Accessibility and Inclusion at oai@lums.edu.pk or the sexual harassment inquiry committee at harassment@lums.edu.pk for any queries, clarifications, or advice. You may choose to file an informal or a formal complaint to put an end of offending behavior. You can find more details regarding the LUMS sexual harassment policy here.

To file a complaint, please write to harassment@lums.edu.pk.

SSE Council on Equity and Belonging

In addition to LUMS resources, SSE's **Council on Belonging and Equity** is committed to devising ways to provide a safe, inclusive and respectful learning environment for students, faculty and staff. To seek counsel related to any issues, please feel free to approach either a member of the council or email at cbe.sse@lums.edu.pk

Rights and Code of Conduct for Online Teaching

A misuse of online modes of communication is unacceptable. TAs and Faculty will seek consent before the recording of live online lectures or tutorials. Please ensure if you do not wish to be recorded during a session to inform the faculty member. Please also ensure that you prioritize formal means of communication (email, LMS) over informal means to communicate with course staff.